

Question #1

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15

- The interrupt vector contains two words (pointers) which specify the address of:
 - Interrupt vector table
 - Interrupt service routine
 - Interrupt type
 - I/O ports
- The interrupt vector for INT 11H is stored in memory at:
 - 44H
 - 4CH
 - 40H
 - 4H
- Assume X string is defined with offset address 100H, find the content of X string and SI register:
 - X='KLM+ORG'; SI=101
 - X='KKK+ORG'; SI=100
 - X='KKKK'; SI=105
 - X='KKK'; SI=100

```
X DB 'KLM+ORG'
MOV CX, 3
MOV SI, OFFSET X
MOV DI, SI
LODSB
L: STOSB
LOOP L
```

- What will be the final value of AX and CX?


```
MOV AX, 15
MOV CX, 4
L1: INC AX
LOOPNZ L1
```

 - AX=13H; CX=0
 - AX=16; CX=3
 - AX=13; CX=0
 - AX=17; CX=0

- What is the value of AL after execution the following codes?


```
CID
X DB "ABCDE"
Y DB "ABHDE"
MOV SI, OFFSET Y
MOV DI, OFFSET X
MOV CX, 5
REPNE CMPSB
LODSB
```

 - AL= 'B'
 - AL= 'D'
 - AL= 'C'
 - AL= 'H'

- The equivalent instructions of STOSB assuming DF flag =0:
 - MOV [DI], AL; INC DI
 - MOV [SI], AL; INC DI
 - MOV AL, [SI]; INC SI
 - MOV [DI], AL; DEC DI

Question #4

A) Fill in the blank spaces with the correct missing components of an 8086 assembly program to accomplish the following
; This section of assembly code sets the cursor position to the coordinates (x, y) = (39, 12).

MOV AH,

MOV BH, 0

MOV DL,

MOV DH,

INT 10h

; This section of assembly code prints \$-terminated string array named string1

MOV AH,

MOV DX,

INT 21h

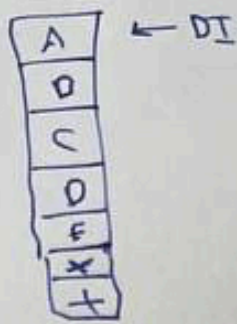
HLT

String1 DB 'ABCDEF\$'

B) Interface 8-bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFh.

a) Draw block circuit diagram circuit, showing address decoding logic, and control signals

b) Write a program to read the status of switches and store the result into X memory location



on 2#: Trace the following programs and determine the final value for each register:

$DI = 0$ $CX = 7$ $AL = 0$
 $SI = A$
 B "ABCDE*+"
 DI, OFFSET STR
 SI, DI
 CX, 7
 AL, 'D'
 SCASB
 SI, DI
 B
 B
 ; CX=
 ; SI= ; STR=

MOV CL, 4
 MOV AX, 20FEH
 TEST AX, 16
 JZ L1
 ROR AL, CL
 SHR AH, CL
 JMP L2
 L1: SHL AL, CL
 ROL AH, CL
 L2: NOT CL

AL= AH=
 CL=

ARRAY DB 4, 2, 7, 9, 4, 5, 5
 STC
 MOV AL, 0
 MOV CX, 4
 MOV AH, 0
 MOV SI, OFFSET ARRAY
 L: ADC AL, [SI]
 ADD SI, 1
 LOOP L
 PUSH AX
 PUSH CX
 POP CX

CX=

X DD 13457ABCH
 MOV DX, X+1
 MOV BX, 00FFH
 PUSH BX
 PUSH DX
 XCHG DH, BH
 POP DX
 POP AX

AX= , BX= ; DX=

on 3#: Write an assembler codes to store numbers from 0 -400 into memory locations
 X DW 400 dup (?)

7. The interrupt vector contains two words (pointers) which specify the address of:
 A. Interrupt vector table B. Interrupt service routine C. Interrupt type D. I/O ports
8. The physical address of interrupt vector associated with INT 4 is:
 A. 00010H B. 00020H C. 00040h D. 00042H
9. An external interrupt breaks the execution of program and diverts it's to execution to:
 A. interrupt service routine B. counter word register C. execution unit D. control unit
10. The size of interrupt vector table is -----
 A. 4 bytes B. 64 bytes C. 1024 bytes D. 128 bytes
11. The binary value of AL and CF after the following instructions have executed is
 MOV CL,4
 MOV AL, 6BH
 ROR AL,CL
 A. AL=11010110B CF=1 B. AL=10110110B CF=0
 C. AL=10110110B CF=1 D. AL=10110111B CF=1
12. What is the binary value of AL after the following instructions have executed?
 MOV AL,C4H
 STC
 SHR AL,1
 A. AL= 62H; CF=0 B. AL= 62H ; CF=1; C. AL=63H; CF=0; D. AL=63H CF=1
13. In the following instruction sequence, show the value of AL, in binary:
 MOV AL,0FH
 NEG AL
 DEC AL
 A. AL= 11110001B B. AL=11110000B C. AL= 00001110B D. AL= 01010000B
14. Given that AL register contains the ASCII code of an uppercase letter, it can be converted to lowercase by
 A. ADD AL, 30B B. AND AL, 0010 0000 C. OR AL, 0010 0000 D. SUB AL, 30
15. How many bits of data can be transferred between the 8255 PPI and the interfaced device at time?
 A. 16 BITS B. 24 BITS C. 8 BITS D. 20 BITS